Digital Government Services for a Greek Municipality

Service prioritisation, target architecture, governance, security, privacy and implementation roadmap

Portfolio case study
Public English edition
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This report develops a practical design for five citizen-facing digital services in a hypothetical medium-sized Greek municipality. It is an educational portfolio artifact and does not use municipal, citizen, or production data.

The public edition removes student identifiers and does not distribute the original university submission.

Executive summary

The recommended programme begins with infrastructure issue reporting and facility booking because both can demonstrate visible resident value without depending on the most sensitive identity, payment, and registry integrations. Digital certificate issuance and municipal debt payments follow after the municipality has tested its architecture, recovery, access control, reconciliation, and operating model. A participation platform forms a later expansion once governance and adoption capacity are established.

The 2025 Digital Decade report describes continued Greek progress in digital public services while identifying gaps in online service coverage and digital skills. It also reports that 78% of Greek respondents believe digitalisation of public and private services makes their lives easier. This creates a strong case for local digital services, but it also makes inclusion, assisted access, and service quality central design requirements (European Commission, 2025).

Recommended sequence

Wave	Services	Decision rationale
Pilot	Issue reporting; facility booking	High visible value with manageable integration and compliance dependencies
Integrate	Certificates; municipal payments	High value, but dependent on trusted identity, records, payment and reconciliation controls
Expand	Digital participation	Outcome depends on governance, moderation, adoption and policy follow-through

Decision conditions

- Redesign each process before moving it online.
- Treat accessibility and assisted service as acceptance criteria.
- Assign a permanent service owner and fund operations after launch.
- Require evidence for privacy, security, recovery, supplier exit and benefits.

Context and case definition

Greek public administration has expanded national digital services through gov.gr and related infrastructure. Local authorities still face uneven maturity, fragmented legacy systems, constrained technical capacity, procurement dependency, and residents with different levels of digital confidence. The national Digital Transformation Bible places citizen-centred digital municipal services, smart-city capabilities, responsible data use, security and confidentiality within the transformation agenda.

Case scope

The case concerns a hypothetical municipality of medium size. It examines five resident-facing services and the shared capabilities needed to deliver them. It does not select products, estimate a procurement budget, or claim measured outcomes from a live deployment.

Design principles

Principle	Application in the case study
User-centred	Design around resident tasks, evidence and exceptions rather than existing organisational boundaries
Digital and assisted	Provide self-service while preserving practical support for residents who need another channel
Once-only where lawful	Reuse verified data through governed interoperability instead of requesting the same evidence repeatedly
Privacy and security by design	Define data, access, logging, retention, recovery and incident requirements before delivery
Open and maintainable	Prefer documented interfaces, portable data and exit arrangements that reduce supplier dependency
Measurable	Baseline the current service and track completion, delay, failure, adoption and support demand

EU interoperability, digital identity, data protection and accessibility instruments provide a common direction. Their application to a specific municipality or service must be confirmed against Greek law and the actual processing and procurement model.

Method and service prioritisation

The analysis combines academic e-government maturity models with official digital-government, legal, accessibility and cybersecurity sources. It uses service scenarios, a weighted comparison, a capability-based architecture, governance mapping and a qualitative risk register. No interviews, survey responses, production data or supplier quotations were available.

Criterion	Weight	Question
Citizen value	30%	How much resident effort, delay or uncertainty could the service remove?
Operational benefit	25%	How much repetitive work, error or queue pressure could it reduce?
Feasibility	20%	Are the technology and dependencies manageable?
Compliance simplicity	10%	How difficult are the privacy, legal and audit obligations?
Delivery simplicity	15%	How manageable are integration, procurement, rollout and support?

Weighted result

Rank	Service	Score	Wave
1	Infrastructure issue reporting	4.28	Pilot
2	Facility booking and appointments	4.10	Pilot
3	Digital certificate issuance	3.95	Integrate
4	Municipal debt payments	3.85	Integrate
5	Digital participation platform	3.35	Expand

The scores are transparent design judgements on a five-point scale. They provide a testable starting hypothesis rather than a universal municipal ranking.

Five service concepts

Service	Resident outcome	Key dependency	Primary control
Infrastructure issue reporting	Submit a geolocated issue, receive a case number and track status	Workflow and field-service ownership	Location and image minimisation; status audit trail
Facility booking and appointments	See availability, reserve a slot and receive confirmation	Calendar and resource ownership	Capacity rules; accessible alternatives; cancellation controls
Digital certificate issuance	Request and receive a verifiable document without unnecessary visits	Trusted identity, registry and signing	Strong identity; document integrity; access logging
Municipal debt payments	View an obligation, pay through a trusted provider and receive confirmation	ERP, payment provider and reconciliation	Hosted payment; idempotency; segregation of duties
Digital participation	Review proposals, submit input and see how the authority responded	Moderation and policy ownership	Transparent rules; privacy controls; public decision trail

Pilot definition

A pilot should cover one clearly bounded resident journey, a limited operational team, real support procedures, monitored accessibility, and a rollback path. It should test whether the municipality can own the service, respond to exceptions and maintain data quality. A polished interface alone does not meet the pilot objective.

Service standard

- Residents can understand eligibility, evidence, status and next steps.
- Staff can manage exceptions without parallel undocumented workarounds.
- The service records decisions and supports review without excessive surveillance.
- Support, security, privacy, accessibility and recovery responsibilities remain clear after launch.

Target architecture

The target separates resident channels from domain services and from legacy systems. Shared identity, API management, observability and audit controls prevent every service from building its own inconsistent foundation.

Layer	Capabilities	Design concern
Channels	Responsive web, mobile access, assisted service	Consistency, accessibility and plain-language guidance
Edge and identity	API gateway, authentication, consent where relevant, abuse protection	Trust boundaries and recovery paths
Domain services	Certificates, issues, payments, bookings, participation	Clear ownership and bounded data
Integration	Workflow, adapters, queues, API contracts	Legacy failure, retries and reconciliation
External systems	Municipal ERP and registries, trusted identity, payment provider, public-sector interoperability	Authority, data quality and supplier responsibility
Data and operations	Operational records, documents, audit logs, monitoring, SIEM, backup	Retention, evidence integrity and recovery

Key architecture decisions

- Use versioned APIs and contract tests instead of direct database coupling.
- Keep payment-card handling with a qualified payment provider through a hosted flow.
- Centralise audit events while limiting who can access the audit trail.
- Apply service-specific retention and deletion rather than one global indefinite archive.
- Design export and supplier exit before signing the delivery contract.

Security, privacy and resilience

Public digital services concentrate identity, financial and case-management data. The design therefore treats trust, availability and accountability as service outcomes. ENISA's public-administration threat landscape reinforces the need to plan for disruptive incidents and data compromise rather than relying only on preventive controls.

Risk	Likelihood	Impact	Required evidence
Identity fraud or account takeover	High	High	IAM configuration; authentication logs
Excessive collection or retention of personal data	Medium	High	Record of processing; retention evidence
Legacy integration failure or inaccurate data exchange	High	High	Interface tests; reconciliation reports
Service outage or destructive cyber incident	Medium	High	Restore tests; incident exercises
Inaccessible service excludes residents	Medium	High	Audit results; user testing
Supplier lock-in and loss of operational knowledge	High	Medium	Exit plan; architecture decision records

Control baseline

- MFA for staff, least privilege, protected account recovery and periodic access review.
- Encryption in transit and at rest with managed keys and documented key ownership.
- Privacy records, lawful basis, minimisation, retention, deletion and DPIA where required.
- Central monitoring, protected logs, tested backup restoration and incident exercises.
- Supplier vulnerability handling, incident notification, audit rights and exit obligations.

Accessibility and inclusive service

The EU Web Accessibility Directive requires public-sector websites and mobile applications to be perceivable, operable, understandable and robust, subject to its scope and national implementation. Accessibility must therefore appear in procurement, design, acceptance and ongoing monitoring rather than as a final visual review.

Acceptance approach

Area	Evidence before launch
Structure and navigation	Keyboard completion, meaningful headings, focus order and visible focus
Forms and errors	Labels, instructions, error identification, recovery and no unnecessary time limits
Documents	Accessible templates, tagged PDFs where used, and an HTML alternative when practical
Content	Plain language, current instructions, contact route and understandable status messages
Assisted service	Staff procedure that can complete the same outcome without penalising the resident
Statement and monitoring	Published accessibility statement, feedback route and remediation ownership

Digital inclusion

A digital channel should reduce avoidable visits without turning non-digital residents into second-class users. The municipality should track assisted-channel demand, abandonment, failed identity checks and recurring support requests. Service research should include older residents, people with disabilities, residents with limited Greek or limited digital skills, and staff who handle exceptions.

Governance and sourcing

A service requires a stable owner who can change policy, process, content, technology and support as evidence changes. Project governance ends at delivery; service governance continues through operation.

Decision	Accountable role	Required evidence
Service outcome and scope	Service owner	Service standard, baseline and prioritised backlog
Data and lawful basis	DPO with service owner	Data map, record of processing and retention schedule
Architecture and integration	IT and architecture lead	Architecture decisions, API contracts and support ownership
Security acceptance	Security lead	Threat model, test evidence, recovery and incident readiness
Accessibility acceptance	Accessibility lead	Audit, user testing, statement and remediation plan
Release and benefits	Service owner	Readiness review, KPI baseline and funded operational plan

Supplier requirements

- Documented interfaces, exportable data and maintained operational documentation.
- Clear security responsibilities, vulnerability handling and incident notification timeframes.
- Accessibility acceptance criteria and remediation obligations.
- Service levels that cover recovery, support and evidence delivery.
- Exit assistance tested before dependency becomes critical.

Implementation roadmap

Phase	Timing	Outcome	Exit evidence
Foundation	Months 0–2	Owners, baselines, architecture, legal and control plan	Approved discovery pack and delivery backlog
Pilot	Months 3–5	Issue reporting and appointment booking for a limited audience	User evidence, accessibility test and operational readiness
Integrate	Months 6–10	Certificate and payment services connected to trusted systems	Reconciliation, recovery and security acceptance
Expand	Months 11–14	Participation service and improvement across the portfolio	Adoption review, funded ownership and benefits report

Delivery gates

- Discovery confirms user need, baseline performance, legal authority and named ownership.
- Design confirms architecture, data, threat, accessibility and supplier responsibilities.
- Pilot confirms staff readiness, assisted access, monitoring, recovery and rollback.
- Scale confirms adoption, control effectiveness, support capacity and funded operation.

The indicative schedule is deliberately conditional. Procurement delays, legacy constraints, national integration dependencies and staff capacity can change the sequence. The municipality should preserve decision gates even when dates move.

Benefits and measurement

Benefits should be measured against a baseline for each service rather than inferred from online availability. A completed build can still increase resident effort if instructions, identity checks, exception handling or support fail.

Outcome	Example KPI	Interpretation
Access	Completion rate by channel	Shows whether residents reach the intended outcome
Speed	Median end-to-end completion time	Includes waiting and back-office handling, not only screen time
Quality	Rework and rejected-request rate	Detects unclear evidence requirements and data defects
Inclusion	Accessibility defects and assisted-channel demand	Reveals barriers and the support capacity still required
Reliability	Availability, failed transactions and recovery performance	Measures operational resilience rather than launch success
Efficiency	Staff handling time and avoidable contact rate	Tests whether process redesign reduced repetitive effort
Trust	Complaints, privacy incidents and satisfaction	Captures confidence and perceived fairness

Illustrative capacity calculation

If a municipality handles 2,000 monthly requests, digitises half of them and saves ten minutes of staff handling per digitised request, the theoretical capacity release is about 167 hours per month. This is an arithmetic scenario, not a measured saving. Real benefits depend on demand, exception rates, channel shift, staff allocation and the cost of operating the digital service.

Limitations and conclusion

Limitations

- The municipality, residents and staff are hypothetical; no primary research was conducted.
- Scores, roadmap dates and the capacity scenario are assumptions that require validation.
- The architecture describes capabilities and boundaries rather than vendor products or a final bill of materials.
- Legal references identify design considerations but do not determine national scope for a specific authority.
- Cybersecurity controls require implementation detail, testing and continuous operation.

Conclusion

A municipality can improve resident service without attempting a single large transformation programme. The stronger path starts with bounded services, shared control foundations and permanent ownership. Issue reporting and booking provide suitable pilots because they test workflow, content, accessibility, support and operational accountability with fewer high-risk integrations. Certificates and payments should follow once identity, interoperability, reconciliation, monitoring and recovery work in practice.

The principal implementation risk is not the absence of software. It is the gap between project delivery and sustainable service operation. Governance, staff participation, data quality, supplier exit, assisted access and measurable outcomes determine whether a digital service remains trustworthy after launch.

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Source and publication note

This public portfolio edition develops an earlier university study by the author. It removes student identifiers, rewrites the work in English, updates the evidence base and adds reproducible prioritisation, governance, architecture, risk and roadmap artifacts. Sources were checked on 16 October 2026.